

01 Evaluation

Critical Assessment of "Rethinking Financial Distress in Public Healthcare"

Detailed Evaluation for Advanced Corporate Finance

EXECUTIVE SUMMARY

Overall Rating: ADEQUATE (C+/B-)

This paper addresses a relevant intersection of corporate finance and public healthcare management by applying Altman's Z"-score model to Greek public hospitals. While it provides valuable empirical documentation of a unique institutional context and identifies an important theoretical puzzle (traditional financial distress models fail in public entities), the study exhibits significant methodological limitations and offers limited theoretical advancement. The paper's primary contribution lies in its detailed empirical description of Greek hospital finances and its call to reconceptualize financial distress for public entities, but it falls short of actually providing that reconceptualization.

Key Strengths:

- Rich empirical documentation of an underexplored context
- Honest acknowledgment that traditional models fail
- Practical relevance to policymakers
- Clear identification of the "continuous bailout" problem

Critical Weaknesses:

- Weak theoretical development and contribution
 - Methodologically limited (single-year, descriptive, atheoretical governance variables)
 - Circular reasoning (uses Z-scores to measure distress, then argues Z-scores are invalid)
 - Limited engagement with Advanced Corporate Finance theory
 - No alternative model proposed despite calling for redefinition
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I. THEORETICAL AND CONCEPTUAL FOUNDATION

Rating: WEAK (2/5)

A. Theoretical Grounding

Assessment: The paper's theoretical foundation is its weakest dimension, falling substantially below the standards expected in Advanced Corporate Finance research. While the paper correctly identifies that traditional financial distress models were developed for private, profit-maximizing firms operating under hard budget constraints with bankruptcy as the ultimate outcome, it fails to develop a coherent alternative theoretical framework for public entities.

Critical Gap 1: Misapplication Without Theoretical Justification

The paper applies Altman's Z"-score—a model explicitly designed to predict bankruptcy probability in private service firms—to a population where bankruptcy is impossible by institutional design. The author states: "the closure of public entities is not a core strategy" and notes that "no bankruptcies or closures of these entities have been reported to date" (p. 2). This creates a fundamental theoretical contradiction that is acknowledged but not resolved. From the perspective of the Advanced Corporate Finance syllabus's coverage of financial distress and bankruptcy (specifically under "Recent Capital Structure Theories and Empirical Tests"), this represents a category error: applying a predictive model where the outcome being predicted (bankruptcy) cannot occur by definition.

The paper attempts to reframe the dependent variable from "probability of bankruptcy" to "probability of mission failure requiring emergency intervention," but this redefinition remains conceptually underdeveloped. What constitutes "mission failure" in a public hospital? Is it inability to provide care, deterioration in clinical quality, workforce exodus, or something else? The paper does not specify clear criteria, making the construct unmeasurable and the theoretical proposition untestable.

Critical Gap 2: Absence of Public Sector Finance Theory

Advanced Corporate Finance theory regarding capital structure (as outlined in the syllabus: "Tradeoff versus pecking order theories; Dynamic Models of CS; Managerial discretion and CS") is built on assumptions of market discipline, access to capital markets, and the trade-off between tax shields and financial distress costs. The paper documents that Greek public hospitals have virtually no long-term debt, no access to bank credit for working capital, and equity that consists largely of government subsidies rather than retained earnings from operations. This fundamentally different financial architecture requires engagement with theories of government ownership, soft budget constraints (Kornai), and the fiscal commons problem—none of which appear in the paper's theoretical framework.

The paper would have been substantially strengthened by engaging with institutional economics literature on state-owned enterprises, public choice theory on budget-maximizing bureaucracies, or even modern corporate finance work on government ownership and privatization (which appears in the syllabus under "International Corporate Finance"). Instead, the theoretical section consists primarily of a literature review of prior studies using Z-scores in healthcare settings.

Critical Gap 3: Agency Theory Superficiality

The paper gestures toward agency theory in its discussion of governance attributes (Section 2.3), noting that "agency theory, which examines the conflicts of interest between principals (e.g., shareholders) and agents (e.g., managers), is a cornerstone of corporate governance studies" (p. 2). However, this engagement remains superficial. The paper does not develop a clear agency-theoretic model of public hospital governance that specifies:

- Who is the principal in a public hospital context? (The state as owner? The Ministry of Health as regulator? Citizens as taxpayers and patients?)
- What are the specific agency conflicts in this setting? (Empire-building? Budget maximization? Rent extraction? Quality shirking?)
- How do the governance mechanisms studied (CEO gender, hospital size, specialty) theoretically relate to these agency conflicts?

From the perspective of the syllabus's coverage of "Corporate Governance (Governance and Equity returns; Board of Directors: Theories and evidence; Executive Compensation)," this treatment is inadequate. The paper treats governance variables as atheoretical controls rather than as theoretically motivated mechanisms for addressing specific agency problems.

Evidence from the Paper:

- "Financial distress requires redefinition in the context of public entities, since their closure is not a core strategy" (p. 1) – acknowledges the problem but provides no theoretical solution
- "This study paves the way for redefining financial distress" (p. 7) – promises a redefinition that is never delivered
- The conceptual framework (Section 2) consists almost entirely of literature review rather than theory development

Connection to Syllabus: The syllabus emphasizes "Theoretical and empirical analyses of corporate financing and investment decisions" and "complex valuations, static and dynamic capital structure." This paper's atheoretical application of a static model to an inappropriate context represents a fundamental misalignment with these learning objectives.

B. Literature Review and Gap Identification

Assessment: The literature review is adequate but narrow, focusing primarily on prior applications of Z-scores to healthcare settings rather than engaging comprehensively with corporate finance theory. The identified gap—that financial distress prediction models fail in public healthcare contexts—is real but not novel.

Strengths:

- Comprehensive coverage of prior work applying Altman's model to hospitals (References 15-17, 31-33, 36)
- Good documentation of financial performance studies in European public hospitals (References 19-22, 34-35)
- Clear articulation of the three main strands of healthcare financial distress research (prediction, retrospective case studies, privatization consequences)

Weaknesses:

The literature review exhibits tunnel vision, focusing almost exclusively on papers that apply Z-scores or similar accounting-based models to healthcare organizations. It fails to engage with several relevant bodies of literature:

1. **Modern Corporate Finance Empirical Methods:** No engagement with contemporary empirical corporate finance methodology (difference-in-differences, regression discontinuity, instrumental variables, natural experiments) as covered in the syllabus under "Empirical Methods." The paper's descriptive, cross-sectional approach seems dated by contemporary standards.
2. **Capital Structure Theory Beyond Z-scores:** The syllabus covers "Dynamic Models of CS; Managerial discretion and CS; Inertial theories: Stock returns and CS; Equity, Market Timing and CS; Adjustment Costs." None of these theoretical frameworks are brought to bear on understanding why public hospitals have the capital structure they do (virtually no debt, equity dominated by subsidies).
3. **International Corporate Finance:** The syllabus includes "Legal origins and Finance; Business Groups and Family firms; International variation in financing and governance patterns." Given that the paper is studying a specific institutional context (Greek public hospitals in a Beveridgean system), engagement with comparative institutional analysis would have been valuable.
4. **Restructuring and Distress Theory:** The syllabus covers topics under corporate control including takeovers and restructuring. The paper could have engaged with this literature to understand public hospital consolidations (mentioned briefly) and restructuring mechanisms in the public sector.

The Identified Gap:

The paper's central claim is that "financial distress remains a relatively underexplored area in public healthcare" and that existing models "fail to account for the support mechanisms that mask poor financial viability, effectively shifting financial distress to key stakeholders such as suppliers and the state owner" (p. 1).

This is presented as a novel finding, but it is not. The literature on soft budget constraints in state-owned enterprises, dating back to Kornai's work in the 1980s, has long established that traditional financial distress concepts do not apply where there is an implicit or explicit government guarantee. The paper's "discovery" that Z-scores overstate financial health in continuously bailed-out public entities is predictable from existing theory, not a genuine research gap.

A more genuine gap would be: "Given that traditional financial distress concepts do not apply to public hospitals, what alternative framework should we use to assess financial sustainability and trigger corrective action?" The paper gestures toward this question but does not answer it.

Evidence from the Paper:

- Literature review (Section 1 and 2.1) cites 53 references, but only 3-4 engage with corporate finance theory directly
- No citations to seminal capital structure papers (Modigliani-Miller, Myers, Titman, etc.) despite the syllabus's emphasis on these foundations
- No engagement with institutional economics or comparative health systems literature that might provide theoretical tools

II. METHODOLOGICAL RIGOR

Rating: **WEAK TO ADEQUATE (2.5/5)**

A. Research Design Appropriateness

Assessment: The research design is fundamentally mismatched to the research question. To assess financial distress—which is by definition a dynamic, temporal process of deterioration—the paper employs a static, single-year cross-sectional snapshot. This creates insurmountable inferential limitations.

Critical Flaw 1: Cross-Sectional Design for a Dynamic Phenomenon

Financial distress is not a static state but a process. Altman's original work and subsequent refinements were explicitly designed to predict whether a firm would experience distress in the next 1-2 years based on current financial ratios. The predictive validity of the model depends on observing which firms subsequently failed and which survived.

This paper applies the Z"-score to 90 Greek public hospitals in 2022, finds all have Z-scores above 5.5 (well above the 2.60 threshold for "financial viability"), and then argues that these scores are misleading because of subsidy dependence and delayed receivables collection. But without temporal data showing which hospitals subsequently experienced "mission failure" (however defined), there is no way to validate or invalidate the Z-score's predictive accuracy in this context.

The author acknowledges this limitation: "This limitation is due to significant lags in financial reporting, as most hospitals have not yet published their financial statements for 2023 and 2024" (p. 5-6). However, this is not merely a data availability problem—it is a fundamental design flaw. Even with 2023-2024 data, the absence of any actual closures or bankruptcies in the population means there is no outcome against which to validate the model.

From the syllabus perspective on "Empirical Methods in Corporate Finance," this design would be considered inadequate. Modern corporate finance research emphasizes identification strategies that allow causal inference, such as difference-in-differences, regression discontinuity, or instrumental variables approaches. The paper's purely descriptive, cross-sectional approach offers no basis for causal claims.

Critical Flaw 2: Hypothesis Testing Without Causal Logic

The paper tests five hypotheses (H1-H5) regarding the relationship between hospital characteristics (location, size, specialty, CEO gender) and Z-scores. However, the research design cannot distinguish between three competing explanations for any observed associations:

1. **Causal effect:** Hospital size causes lower Z-scores (e.g., larger hospitals face diseconomies of scale)
2. **Reverse causality:** Hospitals with lower Z-scores are consolidated into larger units
3. **Omitted variable bias:** A third factor (e.g., case mix complexity) causes both larger size and lower Z-scores

The paper employs only bivariate statistical tests (Mann-Whitney U, Kruskal-Wallis H, Spearman correlations) with no multivariate controls for confounding variables. This is methodologically weak by contemporary standards. The syllabus's emphasis on "empirical analyses of corporate financing and investment decisions" implies the use of regression techniques that control for multiple factors simultaneously.

Critical Flaw 3: No Validation Criterion

In Altman's original work, the Z-score model was validated by showing that firms with low Z-scores had significantly higher bankruptcy rates in subsequent years than firms with high Z-scores. The model's cutoff points ($Z < 1.10$ for distress, $Z > 2.60$ for viability) were calibrated based on observed outcomes in a sample containing both failed and surviving firms.

This paper applies the model to a population with zero bankruptcies. The author correctly notes that "such failures occur globally" (p. 1) and cites examples of hospital closures in Europe and the US, but the studied population (Greek public hospitals, 2022) contains no closures. Without a validation sample containing both "distressed" and "healthy" entities (however defined), there is no empirical basis for assessing whether the Z-score thresholds have any meaning in this context.

The paper attempts to address this by arguing qualitatively that high Z-scores are misleading because they reflect subsidies and delayed receivables, but this is essentially an a priori theoretical argument masquerading as empirical validation.

Evidence from the Paper:

- "This study included 90 public hospitals, representing the total population of such healthcare providers in Greece" (p. 3) – census rather than sample, limiting generalizability
- Single year (2022) data with no longitudinal or panel structure
- "Although no bankruptcies or closures of these entities have been reported to date" (p. 2) – no outcome to validate against
- Statistical analysis (Section 3.4) specifies only "descriptive statistics, normality tests, correlation analysis, and nonparametric tests" – no multivariate regression, no causal identification strategy

Connection to Syllabus: The syllabus emphasizes "Testing CS theories" and empirical methods. The paper's design does not allow for theory testing in any meaningful sense—it is purely descriptive documentation that a particular model yields particular values in a particular setting.

B. Measurement and Data Quality

Assessment: Measurement quality is mixed. Financial statement data appear reliable and the adjustments for accounting standard differences are transparent, but the operationalization of governance variables is atheoretical and the construct validity of Z-score components in this context is questionable.

Strengths:

1. **Transparent Data Source:** All financial data come from published financial statements available through the Ministry of Health portal, providing replicability and auditability.
2. **Thoughtful Accounting Adjustments:** The paper makes explicit adjustments to harmonize local accounting standards (Presidential Decree 146/2003) with IPSAS:
 - Depreciation treatment standardized (treating it as production cost)
 - Irregular income/expenses from prior years excluded to align periods
 - Retained earnings redefined as net income (appropriate given no dividends)

These adjustments are reasonable and enhance comparability across the sample.

3. **Complete Population:** Using all 90 public hospitals eliminates sampling error, though it raises questions about external validity.

Weaknesses:

Measurement Issue 1: Construct Validity of Z-score Components

The paper provides compelling evidence that the Z-score components do not measure what they purport to measure in this context:

- **X1 (Working Capital / Total Assets) as "Liquidity":** The paper finds working capital is 61.7% of total assets on average, seemingly indicating excellent liquidity. However, deeper analysis reveals that 93.5% of current assets are accounts receivable with an average collection period of 1,485 days (over 4 years). The author correctly notes: "This raises the question of how 'current' an asset is if it is converted to cash after 4 years" (p. 5). IPSAS standards classify receivables collected after 12 months as non-current assets. If this classification were applied, most of these "current" assets would be reclassified, dramatically reducing working capital and the X1 ratio. The paper acknowledges this: "X1 overstates actual liquidity by ignoring limited liquid assets, which inflates Z-scores" (p. 5).
- **X4 (Book Equity / Total Liabilities) as "Solvency/Leverage":** The paper finds extraordinarily high equity-to-liability ratios (mean 10.9, median 8.1), suggesting minimal leverage. However, the composition of equity is fundamentally different from private firms:
 - Equity consists primarily of government subsidies and donations, not retained earnings from operations
 - Only 3 of 90 hospitals can self-finance operations without subsidies
 - Liabilities are artificially suppressed because long-term debt is essentially zero (all fixed assets are financed through subsidies)
 The author concludes: "X4 tends to understate financial leverage, leading to inflated Z-scores" (p. 5).

These measurement validity problems are not mere technical details—they fundamentally undermine the paper's central analytical tool. If X1 and X4 (which contribute 11.5 out of the mean 15.7 Z-score) are invalid measures of liquidity and leverage in this context, then the Z-score itself is invalid.

Measurement Issue 2: Atheoretical Governance Variables

The governance variables tested in H2-H5 are measured as:

- **Location:** Categorical variable for 7 Regional Health Authorities
- **Size:** Natural logarithm of total assets and revenue
- **Specialty:** 5-category classification (general, general+health center, university, anticancer, other)
- **CEO Gender:** Binary male/female

These operationalizations are atheoretical convenience measures rather than theory-driven constructs:

- **Location:** No theoretical rationale is provided for why Regional Health Authority boundaries should matter for financial performance. These are administrative divisions that may not correspond to economic regions, population health differences, or healthcare market structures. A more theoretically grounded approach would measure urbanicity, population density, socioeconomic characteristics, or healthcare market concentration.
- **CEO Gender:** The paper treats this as a simple binary variable but provides no theoretical mechanism linking gender to financial distress beyond citing correlational evidence from prior studies. From a corporate governance perspective (covered in the syllabus under "Board of Directors: Theories and evidence; Executive Compensation"), a more sophisticated approach would examine:
 - CEO tenure and whether gender effects differ for new vs. established leaders
 - Interaction between CEO gender and board composition
 - Gender differences in compensation structure and incentive alignment
 - Selection effects (do women get appointed to hospitals that are already in better financial condition?)
- **Specialty:** The classification scheme appears ad hoc. Why are "general hospitals with health centers" a separate category from "general hospitals"? What is the theoretical justification for grouping all "other specialties" together?

Measurement Issue 3: Missing Variables

Several variables that corporate finance theory suggests should be important are not measured:

- **Case Mix Complexity:** Hospitals treating more complex, expensive cases will have different cost structures and financial profiles, but this is not controlled for
- **Teaching Status Intensity:** While university hospitals are flagged, there's no measure of the intensity of teaching and research activities
- **Governance Structure:** Board composition, number of board members, presence of independent directors, frequency of meetings—all standard governance measures are absent
- **Management Compensation:** No data on CEO pay, pay-for-performance structure, or whether managers' incentives are aligned with financial performance
- **Political Connections:** In public entities, political appointment of managers can create agency problems, but this is not measured

Evidence from the Paper:

- "All data were retrieved from published financial statements for 2022" (p. 3) – reliable but limited source
- Detailed accounting adjustments described (Section 3.3) demonstrate methodological care
- "X1 overstates actual liquidity" and "X4 tends to understate financial leverage" (p. 5) – author's own acknowledgment of measurement invalidity
- Governance variables described in Section 3.1 with no theoretical justification for operationalization choices

Connection to Syllabus: The syllabus covers "Corporate Governance" extensively, including board composition, executive compensation, and governance mechanisms. The paper's treatment of governance is superficial by these standards, using crude proxy variables (CEO gender, hospital size) that do not capture the governance mechanisms emphasized in corporate finance theory.

C. Statistical Analysis

Assessment: The statistical analysis is methodologically conservative but limited in scope. The exclusive reliance on bivariate non-parametric tests is appropriate given the data characteristics but sacrifices statistical power and causal inference. The absence of multivariate regression is a significant limitation.

Strengths:

1. **Appropriate Distribution Checks:** The paper conducts Kolmogorov-Smirnov normality tests and finds that most variables are non-normally distributed (Table 1), justifying the use of non-parametric methods.
2. **Correct Non-Parametric Techniques:**
 - Mann-Whitney U test for two-group comparisons (CEO gender)
 - Kruskal-Wallis H test for multi-group comparisons (location, specialty)

- Spearman rank correlations for continuous variables These are appropriate choices given distributional assumptions.

3. **Complete Reporting:** Tables 1-5 provide comprehensive descriptive statistics and test results with all necessary parameters (mean ranks, test statistics, p-values).

Weaknesses:

Statistical Limitation 1: Bivariate Analysis Only

The paper tests each hypothesis using bivariate comparisons without any multivariate controls. This creates severe omitted variable bias problems. For example:

- H3 (hospital size affects Z-scores) finds a negative correlation ($p = -0.266$, $p < 0.05$) between size (measured by log revenue) and Z-scores. But size is likely correlated with other characteristics: larger hospitals are more likely to be urban, more likely to be teaching hospitals, more likely to be general rather than specialized hospitals. Without controlling for these confounders, we cannot isolate the effect of size.
- H5 (CEO gender affects Z-scores) finds female-led hospitals have higher Z-scores ($p = 0.042$), but female CEOs may be systematically different from male CEOs in other ways: they may have different educational backgrounds, tenure, be appointed to hospitals in different stages of their lifecycle, or face different career incentives. A multivariate regression controlling for CEO age, tenure, educational credentials, hospital characteristics, and prior hospital performance would be necessary to isolate gender effects.

Statistical Limitation 2: No Power Analysis or Effect Size Reporting

While the paper reports p-values, it does not report effect sizes or conduct power analysis. For example:

- The Mann-Whitney U test for CEO gender (Table 3) shows $Z = -2.037$, $p = 0.042$. This is statistically significant at the 0.05 level, but barely. With a sample of 69 male-led and 21 female-led hospitals, what is the statistical power to detect a meaningful difference?
- The paper finds no significant effect of specialty on Z-scores ($p = 0.685$, Table 5), accepting the null hypothesis. But with only 2 anticancer hospitals and 6 "other" specialty hospitals in the sample, the test has very low power to detect differences. The null finding may reflect inadequate sample size rather than genuine absence of effect.
- Effect sizes (Cohen's d , rank-biserial correlation for Mann-Whitney) would allow readers to judge practical significance separately from statistical significance.

Statistical Limitation 3: Multiple Hypothesis Testing

The paper tests 5 main hypotheses (H1-H5) plus examines 4 Z-score components for each hypothesis (X1, X2, X3, X4), resulting in approximately 25 statistical tests. No correction for multiple comparisons (e.g., Bonferroni correction) is applied. This inflates the probability of Type I error (false positives). Some of the marginally significant findings (e.g., X4 for CEO gender, $p = 0.06$) would likely become non-significant after multiple testing correction.

Statistical Limitation 4: Limited Exploration of Interactions

The bivariate approach prevents exploration of potentially important interaction effects:

- Does the effect of CEO gender differ by hospital size? (Perhaps female leadership matters more in smaller organizations)
- Does location interact with specialty? (Perhaps university hospitals have different location effects than general hospitals)
- Does size moderate the relationship between leverage and Z-scores?

These interactions could be explored through multivariate regression with interaction terms, but the paper's statistical approach cannot address them.

Statistical Limitation 5: Correlation vs. Causation

Table 2 reports correlations between size measures and Z-score components, finding that larger hospitals have lower Z-scores primarily due to higher leverage (X4: $p = -0.356$, $p < 0.05$). The paper interprets this as "a disadvantage for larger hospitals in terms of financial viability" (p. 4).

However, correlation does not imply causation. Several alternative explanations are consistent with this correlation:

1. Larger hospitals genuinely face diseconomies of scale and require more leverage
2. Hospitals with financial problems are consolidated into larger units (reverse causality)
3. Larger hospitals treat more complex cases, requiring more capital investment (omitted variable)
4. Measurement artifact: larger hospitals may have different accounting practices or more complete reporting of liabilities

Without a research design that addresses endogeneity (e.g., instrumental variables, natural experiment, panel data with fixed effects), these competing explanations cannot be distinguished.

Evidence from the Paper:

- "The research methodology was accomplished using descriptive statistics, normality tests, correlation analysis, and nonparametric tests" (p. 3) – limited statistical toolkit
- No mention of regression analysis, multivariate modeling, or causal identification strategies
- Extensive use of p-values but no effect size reporting: " $p < 0.05$ " appears 15+ times without substantive interpretation
- Tables 2-5 report only bivariate associations, no regression coefficients or controlled estimates

Connection to Syllabus: The syllabus emphasizes "Theoretical and empirical analyses" and "Testing CS theories." The paper's purely descriptive statistical approach falls short of the standards for hypothesis testing in corporate finance. Modern empirical corporate finance (as represented in journals and syllabi) emphasizes:

- Regression analysis with multiple controls
- Addressing endogeneity through identification strategies
- Distinguishing correlation from causation
- Robustness checks and sensitivity analysis

III. EMPIRICAL RESULTS AND INTERPRETATION

Rating: ADEQUATE (3/5)

A. Consistency Between Results and Claims

Assessment: There is a fundamental tension—bordering on circular reasoning—in the paper's interpretation of results. The paper uses Z-scores to measure financial distress, finds all hospitals have high Z-scores indicating financial viability, but then argues that high Z-scores are misleading and mask "true" financial distress. This logical inconsistency undermines the paper's central contribution.

The Circular Reasoning Problem:

The paper's analytical strategy follows this logic:

1. Apply Altman's Z"-score to measure financial distress
2. Find all hospitals have $Z > 5.5$ (mean 15.7), well above the 2.60 threshold for viability
3. Accept H1: "Public hospitals in Greece are not financially distressed"
4. BUT THEN argue that Z-scores are invalid in this context because they reflect subsidies, delayed receivables, and masked problems
5. Conclude: hospitals ARE distressed, just not in the way Z-scores measure

This creates a paradox: if Z-scores are invalid measures of financial distress in public hospitals (which the author convincingly argues in Section 5.2), then why use them as the primary analytical tool? And if they are valid, then why reject their conclusion that hospitals are financially healthy?

The paper seems to want it both ways: accepting the Z-score model's authority when it supports certain hypotheses (H2-H5 about governance attributes) while rejecting it when it yields the "wrong" conclusion (H1 about overall distress levels).

A more logically consistent approach would be:

- **Option A:** Take Z-scores at face value, conclude hospitals are financially healthy, and explain why (state support, soft budget constraints, etc.)
- **Option B:** Reject Z-scores as invalid for this context from the outset, and develop an alternative measure of financial health/distress specific to public hospitals
- **Option C:** Use Z-scores as one imperfect indicator among several, triangulating with other evidence (clinical quality, access, workforce stability, supplier relationships)

The paper attempts something like Option C in Section 5.2 by examining "factors beyond Z-scores," but this analysis is descriptive rather than systematic, and no alternative quantitative measure is developed.

Interpretation Issues with Specific Findings:

H1 (Financial Distress):

- **Finding:** All 90 hospitals have $Z > 5.5$, mean $Z = 15.7 \pm 10.8$
- **Paper's Interpretation:** "H1 is accepted, as no hospital was classified as financially distressed... However, this result warrants a more in-depth analysis" (p. 4)
- **Critical Assessment:** The "however" undermines the acceptance of H1. The paper provides extensive qualitative evidence of financial stress:
 - €1.9 billion in state subsidies needed for operations (only 3 hospitals self-financing)
 - 1,485-day average collection period for receivables
 - Rising overdue liabilities (€344M in 2019 → €907M in 2022 → €1,164M in 2024)
 - Heavy reliance on clawback mechanisms

This evidence suggests that hospitals ARE financially distressed by any reasonable definition, but the Z-score model fails to capture it. The paper acknowledges this: "financial condition substantially weaker than that implied by Z-scores" (p. 1). So should H1 be accepted or rejected? The paper's position is unclear.

H3 (Hospital Size):

- **Finding:** Smaller hospitals have significantly higher Z-scores ($p = -0.266$, $p < 0.05$ between log revenue and Z-score)
- **Paper's Interpretation:** "This suggests a disadvantage for larger hospitals in terms of financial viability" (p. 4)
- **Critical Assessment:** This interpretation contradicts the paper's own argument that high Z-scores do not indicate financial viability. If high Z-scores are misleading (because they reflect accounting artifacts rather than genuine financial health), then the finding that small hospitals have higher Z-scores does not mean they are actually in better financial condition—it might mean their accounting profiles happen to fit the Z-score formula better.

The paper notes that smaller hospitals have lower leverage (X4), which mechanically increases Z-scores, but doesn't explore whether this reflects genuine financial strength or simply different financing patterns. Small rural hospitals might have lower leverage because they lack access to capital markets and must rely entirely on government grants, which would indicate weakness rather than strength.

H5 (CEO Gender):

- **Finding:** Female-led hospitals have higher Z-scores (mean rank 55.67 vs. 42.41, $p = 0.042$)
- **Paper's Interpretation:** "Hospitals with female CEOs achieved higher Z-scores, primarily driven by better solvency ratios. This result aligns with prior studies indicating that women in C-suite positions contribute to reducing organizational leverage" (p. 4-5)
- **Critical Assessment:** This interpretation suffers from two problems:
 1. **Endogeneity/Selection:** The paper assumes CEO gender causes financial performance, but the causal arrow could run the opposite direction. Perhaps women are appointed to hospitals that are already in better financial condition, or women systematically sort into easier-to-manage hospitals. Without

panel data showing financial performance before and after CEO transitions, or an exogenous source of variation in CEO gender (e.g., government policy mandating gender diversity), this is purely correlational.

2. **Mechanism Unclear:** The paper cites evidence that female executives adopt "less risky and less aggressive financial strategies" (p. 5), but this doesn't obviously apply to public hospitals where managers don't choose their capital structure. The paper shows female-led hospitals have higher equity-to-liability ratios, but this likely reflects government subsidy allocations rather than managerial choices. The mechanism linking CEO gender to financial ratios in a public entity with no control over financing remains unexplained.

H2 (Location) and H4 (Specialty):

- **Findings:** Location affects individual Z-score components (X1, X2, X3) but not overall Z-score; Specialty has no effect
- **Paper's Interpretation:** "H2 is partially supported... H4 is rejected" (p. 4)
- **Critical Assessment:** These null findings for the overall Z-score, combined with significant effects on components, suggest that the Z-score's weighting scheme may be inappropriate for this context. Altman's original weights (6.56, 3.26, 6.72, 1.05 for X1-X4) were calibrated using discriminant analysis on a sample of manufacturing firms. There's no reason to expect these weights to be optimal for public hospitals, where the relative importance of liquidity, profitability, efficiency, and leverage may be completely different.

Evidence from the Paper:

- "H1 is accepted" but "this result warrants a more in-depth analysis" (p. 4) – hedged conclusion
- "X1 overstates actual liquidity" and "X4 tends to understate financial leverage, leading to inflated Z-scores" (p. 5) – author's own critique of the measure
- Discussion section (5.1) both accepts and rejects findings simultaneously
- "Financial distress requires redefinition" (p. 1, 7) but no redefinition is provided

B. Discussion of Anomalous Findings

Assessment: The paper acknowledges several findings that contradict prior literature or expectations but provides limited theoretical explanation for these anomalies. This represents a missed opportunity for theoretical development.

Anomaly 1: Smaller Hospitals Outperform Larger Ones

- **The Puzzle:** The paper finds smaller hospitals have higher Z-scores, contradicting prior research by Mosciaro et al. and Langabeer et al. showing smaller facilities are in poorer financial condition, and contradicting basic economic theory about economies of scale.
- **Paper's Explanation:** "This outcome reflects lower leverage, despite lower profitability and efficiency, compared to larger hospitals" (p. 4)
- **Critique of Explanation:** This is a description of the statistical pattern, not an explanation. WHY do smaller hospitals have lower leverage? Possible theoretical explanations could include:
 - **Capital access:** Smaller hospitals can't access debt markets and must rely entirely on government grants, appearing "conservative" in leverage ratios but actually indicating financial constraints
 - **Risk selection:** Larger hospitals treat more complex, capital-intensive cases requiring more equipment and facilities, necessitating higher leverage
 - **Scale economies:** Larger hospitals can use leverage more efficiently due to economies of scale in debt servicing
 - **Accounting artifacts:** Smaller hospitals may have older, fully depreciated assets leading to lower asset bases and mechanically higher equity-to-asset ratios

The paper does not adjudicate between these competing explanations, leaving the anomaly unresolved.

Anomaly 2: Rural Hospitals Outperform Urban Ones

- **The Puzzle:** The paper finds hospitals in predominantly rural RHAs (3rd, 4th, 7th) outperform urban hospitals on liquidity, profitability, and efficiency, contradicting prior literature (Malone et al., Holmes et al.) showing rural hospitals are more vulnerable.
- **Paper's Explanation:** "Although these findings are not directly comparable to those of Malone et al. (37) and Holmes et al. (15), they do not support the claim that rural hospitals are particularly vulnerable to financial distress" (p. 4)
- **Critique of Explanation:** The paper dismisses the contradiction without explaining it. Possible explanations could include:
 - **Context specificity:** Greek rural hospitals face different competitive dynamics than US rural hospitals (the comparison cases in the cited studies)
 - **Subsidy allocation:** Greek government may allocate proportionally more subsidies to rural hospitals to maintain access
 - **Case mix:** Rural hospitals in Greece may handle simpler, less capital-intensive cases, improving their financial ratios
 - **Urban hospital problems:** Urban hospitals may face specific challenges (union strength, political interference, more complex governance) that outweigh their advantages

Again, the paper identifies the pattern but provides no theoretical account.

Anomaly 3: Specialty Doesn't Matter

- **The Puzzle:** H4 is rejected; hospital specialty has no significant effect on Z-scores or components, contradicting Khullar et al. who found teaching hospitals have better liquidity.
- **Paper's Explanation:** "This contradicts Khullar et al. (38), who suggested that hospitals with higher liquidity are more likely to be university hospitals" (p. 4)
- **Critique:** This is the paper's weakest discussion of an anomalous finding—it simply notes the contradiction and moves on. No explanation is attempted. Possible theoretical accounts:
 - **Standardized financing:** In a state-controlled system, all hospital types may receive similar subsidy formulas, homogenizing their financial profiles regardless of specialty
 - **Sample size:** With only 7 university hospitals, 2 anticancer hospitals, and 6 "other" in the sample, statistical power may be insufficient to detect real differences
 - **Measurement:** The specialty categories may not capture the relevant distinctions (e.g., research intensity, teaching intensity, case complexity)

The paper's failure to explore this null finding is particularly problematic because the syllabus covers "Corporate Diversification (Conglomerates; The diversification discount Controversy)" which suggests tools for analyzing how organizational scope and specialization affect financial performance.

Anomaly 4: Profitability and Efficiency Are Not Related to Overall Z-scores

- **The Puzzle:** Table 2 shows that X2 (profitability) and X3 (efficiency) are not significantly correlated with overall Z-scores, while X4 (leverage) is strongly correlated ($p = 0.90+$, not reported but implied by the discussion).
- **Paper's Explanation:** None provided.
- **Critique:** This suggests the Z-score in this context is almost entirely determined by leverage (X4), with the profitability and efficiency components contributing little to variation across hospitals. This raises the question of whether a simpler leverage-based measure would be more appropriate than the complex Z-score formula.

Evidence from the Paper:

- Size effect discussed (p. 4) but mechanism unexplained
- Rural vs. urban pattern noted (p. 4) but dismissed rather than explained
- Specialty null finding mentioned (p. 4) but not explored
- Component vs. overall score disconnect not discussed despite being evident in tables

Connection to Syllabus: The syllabus covers "Recent Capital Structure Theories and Empirical Tests" which would provide tools to explain the leverage patterns observed (e.g., pecking order theory might explain why hospitals rely on government subsidies rather than debt). The paper misses this opportunity for theoretical engagement.

IV. CONTRIBUTION TO KNOWLEDGE

Rating: **WEAK TO ADEQUATE (2.5/5)**

A. Theoretical Contribution

Assessment: The paper makes a limited theoretical contribution. While it identifies an important problem (traditional financial distress concepts don't apply to public entities), it does not develop an alternative theoretical framework. The paper is primarily descriptive and diagnostic rather than theoretically generative.

Claimed Theoretical Contribution:

The paper claims to "pave the way for redefining financial distress in public healthcare organizations" and argues that "financial distress requires redefinition in the context of public entities, since their closure is not a core strategy" (p. 7).

Evaluation of Claimed Contribution:

This is more of a research agenda than an actual contribution. The paper:

- Identifies the problem: existing models don't work
- Demonstrates empirically: Z-scores are high despite obvious financial stress
- Calls for: a new definition and model
- BUT does not provide: that new definition or model

From a corporate finance perspective, a genuine theoretical contribution would involve:

1. **Conceptual Development:** Clearly defining what "financial distress" means in a soft budget constraint environment where bankruptcy is impossible. The paper suggests "mission failure requiring emergency intervention" but never operationalizes this.
2. **Model Building:** Developing a formal model (mathematical or conceptual) that predicts when public entities will experience financial crisis, incorporating unique features like subsidy dependence, political constraints, and stakeholder costs.
3. **Testable Hypotheses:** Deriving predictions from the new model that differ from traditional financial distress theory and can be empirically tested.

The paper does none of these. It stops at problem identification.

Comparison to Syllabus Standards:

The Advanced Corporate Finance syllabus emphasizes "Theoretical and empirical analyses of corporate financing and investment decisions" and covers "Recent Capital Structure Theories." A paper at this level should engage with existing theory at a sophisticated level and extend it in novel directions.

For example, a strong theoretical contribution might:

- Extend trade-off theory to incorporate soft budget constraints and government guarantees
- Develop a multi-period game-theoretic model of public hospital financing with state as principal and hospital managers as agents
- Apply real options theory to model the government's option to restructure or close a hospital
- Extend pecking order theory to explain financing choices in state-owned enterprises

The paper cites these theories but does not develop them.

What the Paper Actually Contributes Theoretically:

The most valuable theoretical insight is the identification of what might be called the **"transferred distress"** phenomenon: public entities don't experience financial distress themselves (due to government support), but they transfer that distress to other stakeholders—suppliers through delayed payments, taxpayers through subsidies, and patients through quality deterioration.

This is expressed in the conclusion: "financial distress in public healthcare organizations... would only be uncovered retrospectively, even when all warning signs are present" and entities "may be considered inherently financially distressed once they become entirely dependent on government funding in perpetuity, while simultaneously imposing financial pressure on their suppliers" (p. 7).

This is a valuable conceptual insight, but it remains underdeveloped. A fuller theoretical treatment would:

- Model the distribution of distress costs across stakeholders
- Identify the equilibrium level of "transferred distress"
- Predict when this arrangement becomes unsustainable
- Specify the triggers for government intervention

Evidence from the Paper:

- Abstract claims: "Financial distress requires redefinition" (p. 1)
- Conclusion: "This study paves the way for redefining financial distress" (p. 7)
- No formal model, no new theory, no alternative framework actually presented
- Conceptual framework (Section 2) is literature review, not theory development

B. Empirical Contribution and Generalizability

Assessment: The empirical contribution is stronger than the theoretical contribution but still limited. The paper provides valuable documentation of Greek public hospital finances, but the single-country, single-year design limits generalizability.

Empirical Contributions:

1. **Comprehensive Documentation:** The paper provides complete coverage of Greek public hospitals (N=90, entire population) with detailed financial data for 2022. This is valuable for researchers and policymakers interested in the Greek healthcare system specifically.
2. **Novel Findings About Greek Context:**
 - Quantification of subsidy dependence: €1.9B in subsidies for €336M net income
 - Extreme receivables collection lag: 1,485 days average
 - Capital structure uniqueness: equity-to-liability ratio of 10.9:1
 - Gender effects in public hospital leadership
 - Geographic variation in financial performance
3. **Replication of Known Patterns:** The paper confirms in the Greek context patterns observed elsewhere:
 - Rising overdue liabilities (found in Portugal, Poland)
 - Financial pressure on suppliers through delayed payments
 - Soft budget constraint problems in state-owned entities

Limitations on Generalizability:

Geographic/Institutional Specificity: The findings are deeply embedded in the Greek institutional context:

- Legacy of the 2008-2015 economic crisis and bailout programs
- Unified sickness fund (EOPYY) financing structure
- "Clawback" mechanism where suppliers repay excess sales
- Specific regional health authority organization
- Presidential Decree 146/2003 accounting standards

How much of this transfers to other Beveridgean systems (NHS England, SNS Portugal, Nordic countries) is unclear. The paper makes no systematic attempt to assess external validity.

Temporal Specificity: The paper studies a single year (2022), which may be atypical:

- Post-COVID period with unusual financial flows
- Specific political context
- Point in time in the accumulation of overdue liabilities
- Specific cohort of hospital managers

Longitudinal data would show whether patterns are stable or transient.

Limited Novel Methodological Contribution: The paper applies an existing model (Altman's Z"-score) to a new context but does not develop new methods or analytical techniques. It does not propose an improved model despite identifying the existing model's inadequacies.

What Would Constitute a Stronger Empirical Contribution:

1. **Comparative Design:** Comparing Greek public hospitals to:
 - Private hospitals in Greece (market discipline effect)
 - Public hospitals in other countries (institutional variation)
 - Greek hospitals before/after major reforms (temporal variation)
2. **Alternative Outcomes:** Measuring actual indicators of financial crisis:
 - Emergency government interventions
 - Forced mergers or restructurings
 - Service closures or reductions
 - Staff strikes or exodus
 - Supplier litigation
3. **Validation:** Showing that some measurable characteristic (could be a modified Z-score or alternative metric) actually predicts which hospitals subsequently experience crisis

Evidence from the Paper:

- Comprehensive Greek data: 90 hospitals, complete financial statements, multiple governance attributes
- Single country, single year: "This study included 90 public hospitals, representing the total population of such healthcare providers in Greece" (p. 3)
- No comparative or longitudinal design
- No proposal for alternative measurement approach despite critique of Z-scores

Connection to Syllabus: The syllabus covers "International Corporate Finance (Legal origins and Finance; Business Groups and Family firms; International variation in financing and governance patterns)." The paper studies one country but doesn't situate it in comparative perspective, missing an opportunity to contribute to understanding cross-national variation in public hospital finance.

V. CORPORATE FINANCE COURSE ALIGNMENT

Rating: ADEQUATE (3/5)

Assessment of Alignment with Syllabus Objectives

The Advanced Corporate Finance syllabus specifies that the course "develops the 'art and science' of optimal strategic decision-making by applying corporate financial theory to cases of financial policy, financial instruments, and valuation." It covers five main content areas:

1. Recent Capital Structure Theories and Empirical Tests
2. Corporate Governance
3. The Market for Corporate Control
4. International Corporate Finance
5. Corporate Diversification and Real Options

The paper engages meaningfully with areas 1 and 2, tangentially with 4, and not at all with 3 and 5.

A. Connection to Specific Syllabus Topics

Topic 1: Capital Structure (MODERATE ENGAGEMENT)

Syllabus coverage: "Tradeoff versus pecking order theories; Dynamic Models of CS; Managerial discretion and CS; Testing CS theories; Inertial theories: Stock returns and CS; Equity, Market Timing and CS; Adjustment Costs"

Paper's treatment: The paper documents an extreme capital structure (virtually zero long-term debt, equity-to-liability ratio of 10.9:1) but does not engage with capital structure theory to explain it.

Missed opportunity: The paper could have used **pecking order theory** to explain why public hospitals rely entirely on government subsidies (analogous to retained earnings) and never use debt or external equity. This theory predicts that firms prefer internal financing to external financing due to information asymmetries. For public hospitals, the "internal" financing is government subsidies (from the owner's perspective), while external financing (bond issues, bank loans) would subject them to market discipline and information disclosure requirements that might be politically inconvenient.

Missed opportunity: **Trade-off theory** could explain the observed capital structure as an optimal balance between the tax shield benefits of debt (zero for public entities that don't pay taxes) and financial distress costs (very high for hospitals because distress threatens their public mission). This predicts zero debt, consistent with observations.

The paper mentions these theories in passing when discussing the project documents, but doesn't apply them.

Rating for CS coverage: 2/5 (describes capital structure but doesn't explain it using CS theory)

Topic 2: Corporate Governance (MODERATE ENGAGEMENT)

Syllabus coverage: "Governance and Equity returns; Board of Directors: Theories and evidence; Executive Compensation; Market overvaluation and Corporate Fraud"

Paper's treatment: The paper examines governance attributes (CEO gender, hospital size as proxy for governance complexity) and their relationship to financial performance (measured by Z-scores). This directly addresses governance effects on financial outcomes.

Strength: The CEO gender analysis engages with corporate governance literature on diversity and financial conservatism, finding female-led hospitals have higher Z-scores driven by lower leverage.

Weakness: The governance analysis is superficial. The paper does not examine:

- Board composition (size, independence, expertise)
- Executive compensation structure or incentives
- Agency conflicts specific to public entities (manager-government relationship)
- Governance mechanisms for monitoring and control

The syllabus specifically covers "Board of Directors: Theories and evidence" and "Executive Compensation," but the paper has no data on boards or compensation.

Rating for Governance coverage: 3/5 (addresses governance-performance link but with limited governance measures)

Topic 3: Market for Corporate Control (NO ENGAGEMENT)

Syllabus coverage: "Takeover models; Stock returns and acquisitions; Mergers and inertial investors; Recent Evidence on takeovers"

Paper's treatment: The paper briefly mentions that "some public hospitals were unified" and uses consolidated vs. individual financial statements, but there is no analysis of hospital mergers, consolidations, or restructurings.

Massive missed opportunity: The paper could have analyzed:

- Greek hospital consolidations as a form of government-mandated merger
- Financial characteristics of hospitals pre/post-merger
- Whether mergers improve or worsen financial performance
- Whether distressed hospitals are targets for consolidation

This would have been directly relevant to the syllabus's coverage of M&A and provided novel evidence on public sector consolidations.

Rating for Corporate Control coverage: 0/5 (topic not addressed)

Topic 4: International Corporate Finance (WEAK ENGAGEMENT)

Syllabus coverage: "Legal origins and Finance; Business Groups and Family firms; International variation in financing and governance patterns"

Paper's treatment: The paper studies a specific institutional context (Greek public hospitals in a Beveridgean system) but does not situate this in comparative perspective.

Partial engagement: The paper acknowledges that "international comparisons of health systems are often confounded by fundamental differences in national definitions" and focuses on one system, which is methodologically sound.

Missed opportunity: The paper could have:

- Compared Greek patterns to other Beveridgean systems (NHS, SNS, Nordic)
- Analyzed how legal/institutional features shape financial architecture
- Used Greece's unique characteristics (economic crisis, bailout programs, clawback mechanisms) as a natural experiment

Rating for International Finance coverage: 2/5 (context-specific study without comparative analysis)

Topic 5: Corporate Diversification and Real Options (NO ENGAGEMENT)

Syllabus coverage: "Conglomerates; The diversification discount Controversy; Real options, types and investment decision"

Paper's treatment: No engagement with diversification or real options despite these being highly relevant to the context.

Missed opportunity - Diversification: Hospital specialty (general vs. university vs. anticancer) could be analyzed through diversification theory. Do hospitals providing a broader range of services (general hospitals) have different financial profiles than specialized hospitals? The paper finds no effect of specialty (H4 rejected) but doesn't interpret this through diversification theory.

Massive missed opportunity - Real Options: Real options theory is directly relevant to government decisions about public hospitals:

- **Option to restructure:** Government has an option to force hospital merger/consolidation
- **Option to privatize:** Could sell or contract out services if performance deteriorates
- **Option to close:** Ultimate option, rarely exercised but always available
- **Option to expand:** Could invest in new facilities or services

The value of these options affects optimal government policy. The paper could have modeled the government's decision as a real options problem, consistent with the Gemini and ChatGPT project documents' emphasis on real options for digital health investments.

Rating for Diversification/Real Options coverage: 0/5 (topics not addressed)

B. Methodological Sophistication Relative to Course Standards

Syllabus expectation: "Theoretical and empirical analyses of corporate financing and investment decisions... advanced topics in corporate finance"

Paper's methodology: Descriptive statistics + bivariate non-parametric tests

Gap: Modern empirical corporate finance uses:

- Multivariate regression with controls
- Panel data methods with fixed effects
- Identification strategies (DiD, IV, RDD)
- Structural modeling
- Machine learning for prediction

The paper uses none of these advanced techniques. Its methodological approach is more consistent with 1980s-1990s empirical finance than contemporary standards.

However: The paper is published in a public health journal (*Frontiers in Public Health*), not a finance journal. By public health standards, the methodology may be adequate. The question is whether it meets standards for an Advanced Corporate Finance course.

Rating for methodological sophistication: 2/5 (appropriate for health management, dated for finance)

Overall Course Alignment Assessment

Strengths:

- Addresses governance-performance relationship (central to course)
- Examines capital structure empirically
- Provides institutional detail on a non-standard organizational form
- Could stimulate class discussion on limits of corporate finance tools

Weaknesses:

- Limited engagement with capital structure theory
- Superficial governance analysis
- No coverage of 2-3 major syllabus topics
- Methodologically dated by finance standards
- Misses opportunities to apply real options, diversification theory

Pedagogical value: This paper would be useful in class as:

1. A case study of applying (and misapplying) finance tools to non-corporate contexts
2. A discussion starter on institutional differences and model validity
3. An example of how institutional context shapes capital structure
4. A demonstration of research design challenges

Rating: 3/5 - Adequate alignment with course, valuable for discussion, but doesn't represent best practice in corporate finance research

VI. PRACTICAL AND POLICY IMPLICATIONS

Rating: ADEQUATE TO GOOD (3.5/5)

A. Actionability of Recommendations

Assessment: The paper provides several policy recommendations, some actionable and some vague. The quality of recommendations is uneven, with some well-grounded in findings and others speculative.

Recommendation 1: Promote Gender Diversity in Hospital Leadership

Paper's recommendation: "Promoting gender diversity in hospital leadership could enhance financial stability. Policymakers and healthcare administrators may consider initiatives to support the recruitment, retention, and advancement of women into executive positions" (p. 5).

Evaluation:

- **Evidence basis:** Finding that female-led hospitals have higher Z-scores ($p = 0.042$)
- **Causal mechanism:** Paper cites research on female executives' risk aversion and financial conservatism
- **Actionability:** Specific (gender diversity targets), feasible (recruitment/promotion policies)
- **Concerns:**
 - Causality uncertain (selection effects possible)
 - Effect size modest (marginally significant p-value)
 - Unclear whether high Z-scores indicate genuine strength (given paper's critique of Z-scores)
 - Risk of oversimplification (gender is proxy for unmeasured characteristics)

Actionability rating: 3/5 - Specific enough to implement but rests on potentially fragile empirical foundation

Recommendation 2: Tailor Financial Oversight by Hospital Size

Paper's recommendation: "Policymakers should tailor financial oversight and support based on hospital size. Larger hospitals, with higher leverage and lower Z-scores, may require targeted interventions to manage debt and enhance financial stability, while best practices from smaller, more resilient hospitals could inform efficiency and liquidity improvements" (p. 4).

Evaluation:

- **Evidence basis:** Negative correlation between size and Z-scores
- **Actionability:** Relatively specific (size-differentiated oversight)
- **Concerns:**
 - Assumes smaller hospitals are "more resilient" based on higher Z-scores, contradicting paper's argument that Z-scores are misleading
 - No specification of what "targeted interventions" would involve
 - "Best practices" from small hospitals may not transfer to large hospitals (scale differences)

Actionability rating: 2.5/5 - Direction clear but implementation details lacking

Recommendation 3: Address Geographic Disparities

Paper's recommendation: "The variation in financial performance across geographic areas suggests that policymakers and central administration should develop mechanisms to monitor and address these discrepancies" (p. 4).

Evaluation:

- **Evidence basis:** Significant variation in X1-X3 components by region
- **Actionability:** Very vague - no specific mechanisms proposed
- **Concerns:**
 - Is geographic variation a problem or appropriate adaptation to local conditions?
 - Should policy aim for geographic uniformity or accept differentiation?
 - No specification of how to "address" discrepancies

Actionability rating: 1.5/5 - Too vague to guide action

Recommendation 4: Reform Financial Distress Measurement

Paper's recommendation (implicit): "Financial distress requires redefinition in the context of public entities... This implies that such supporting mechanisms should be transparently disclosed in financial reports" (p. 7).

Evaluation:

- **Evidence basis:** Strong (Z-scores don't capture subsidy dependence, receivables lag, transferred distress)
- **Actionability:** General direction (disclosure) but no specific metrics proposed
- **Value:** High - identifies important gap in current financial reporting
- **Concerns:**
 - Paper doesn't specify what should be disclosed beyond current reporting
 - No alternative financial distress model proposed despite calling for redefinition

Actionability rating: 2/5 - Identifies problem but doesn't solve it

Recommendation 5: Monitor and Address Payment Delays

Paper's recommendation (implicit in discussion): "Delayed payments primarily affect smaller local suppliers, undermining competition. Larger suppliers may charge interest or refuse to supply, worsening liquidity in a vicious cycle" (p. 5).

Evaluation:

- **Evidence basis:** Strong (1,485-day receivables, €1,164M overdue liabilities by 2024)
- **Actionability:** High - payment discipline is directly policy-controllable
- **Value:** High - addresses real operational problem with stakeholder impacts
- **Specific actions:** Could include payment term enforcement, supplier financing programs, EOPY processing improvements

Actionability rating: 4/5 - Clear problem, measurable metrics, obvious policy levers

B. Practical Value for Different Stakeholders

For Hospital Managers:

- **Value:** Limited. The Z-score analysis doesn't provide actionable management insights
- **Useful findings:** Gender diversity may have financial benefits; size matters for financial management
- **Missing:** Specific financial management strategies, operational improvements, negotiating strategies with government funders

For Policymakers/Regulators:

- **Value:** Moderate. Identifies systemic problems (payment delays, subsidy dependence) requiring policy attention
- **Useful findings:** Traditional financial oversight tools don't work; need new metrics
- **Missing:** Specific policy designs, cost-benefit analysis of interventions, comparative evidence from other countries' solutions

For Researchers:

- **Value:** Moderate. Identifies a research agenda (develop new financial distress concepts for public entities) but doesn't execute it
- **Useful:** Comprehensive data on Greek hospitals; clear statement of theoretical puzzle
- **Missing:** Theoretical development, methodological innovation, replicable framework

Overall Practical Implications Rating: 3.5/5 The paper identifies real problems and provides directionally correct recommendations, but lacks the specificity and causal grounding needed for confident policy action.

VII. LIMITATIONS AND FUTURE RESEARCH

Rating: ADEQUATE (3/5)

A. Acknowledgment of Limitations

Assessment: The paper acknowledges some limitations honestly but understates others and misses several important ones.

Acknowledged Limitations:

1. Single-Year Design (Well Acknowledged): "The main limitation of this study is that it covers a single fiscal year (2022) and, therefore, cannot capture trends over time" (p. 5).

Evaluation: This is appropriately flagged and explained (reporting lags, COVID effects). However, the paper understates how much this limitation undermines its conclusions.

2. Absence of Bankruptcies (Acknowledged but Minimized): The paper notes "no bankruptcies or closures of these entities have been reported to date" (p. 2) but doesn't emphasize that this makes the Z-score model fundamentally unvalidatable in this context.

Unacknowledged or Understated Limitations:

1. Construct Validity (Partially Acknowledged): The paper extensively discusses that X1 and X4 don't measure what they're supposed to in this context (Section 5.2), but this is presented in the discussion rather than the limitations section. The fundamental problem—that the primary analytical tool is invalid for the research context—should be prominently acknowledged as a core limitation.

2. Causal Inference (Not Acknowledged): The paper tests hypotheses about relationships between governance attributes and financial performance (H2-H5) using cross-sectional correlations, but never acknowledges that causal conclusions cannot be drawn from this design. Endogeneity, selection bias, and omitted variable problems are not mentioned.

3. Generalizability (Not Acknowledged): The paper studies Greek public hospitals in 2022—a specific institutional context during a specific historical moment (post-COVID, post-crisis). No discussion of how findings might transfer to:

- Other countries' public hospitals
- Other types of public entities (schools, municipalities)
- Greek private hospitals
- Different time periods

4. Measurement of Governance Variables (Not Acknowledged): The use of CEO gender, hospital size, and specialty as proxies for governance without measuring actual governance mechanisms (boards, compensation, oversight) is not acknowledged as a limitation.

5. Statistical Power (Not Acknowledged): With small subgroups (21 female CEOs, 2 anticancer hospitals), some null findings may reflect inadequate statistical power rather than genuine absence of effects.

6. Circular Reasoning (Not Acknowledged): The logical inconsistency of using Z-scores to measure distress while arguing Z-scores are invalid is not acknowledged.

Evidence from the Paper:

- Limitations section (5.3) is brief: 2 paragraphs addressing only temporal and reporting issues
- No discussion of methodological limitations
- No acknowledgment of theoretical limitations
- Construct validity problems discussed in Section 5.2 but not flagged as limitations

B. Future Research Agenda

Assessment: The proposed future research is directionally appropriate but lacks specificity and feasibility assessment.

Proposed Future Research:

1. "Validate Z-scores over time and refine them to capture subsidy dependence, delayed cash inflows, and other structural vulnerabilities" (p. 6)

Evaluation:

- **Value:** High - addresses core limitation
- **Specificity:** Low - doesn't specify how to "refine" or what the refined model would look like
- **Feasibility:** Uncertain - requires longitudinal data and agreed-upon crisis indicators

2. Implied: Develop alternative financial distress definition

The paper calls for "redefining financial distress" but doesn't propose a concrete research program.

Better specification would be:

- What outcomes constitute "mission failure" or "crisis" in public hospitals?
- What financial and operational indicators predict these outcomes?
- How can we validate a new model in the absence of bankruptcies?

What's Missing from the Future Research Agenda:

1. Comparative Research:

- Cross-country comparison of public hospital finances
- Comparison to private hospitals in mixed systems
- Natural experiments from health reforms

2. Causal Identification:

- Panel data analysis with hospital fixed effects
- Difference-in-differences studies of policy changes
- Instrumental variables for governance attributes

3. Alternative Outcomes:

- Link financial metrics to clinical quality indicators
- Study relationship between finances and patient access
- Examine workforce outcomes (retention, strikes, satisfaction)

4. Mechanism Research:

- How do soft budget constraints affect managerial incentives?
- What triggers government intervention in distressed hospitals?
- How do political cycles affect hospital finances?

5. Solution Development:

- Design and test alternative performance metrics for public hospitals
- Develop decision support tools for government funders
- Propose contractual structures that align incentives

From Syllabus Perspective:

The Advanced Corporate Finance syllabus emphasizes "Empirical Methods" and "Testing CS theories." A strong future research section would specify:

- Identification strategies for establishing causality
- Data requirements and sources
- Specific hypotheses derived from theory
- Expected contribution to corporate finance knowledge

Overall Limitations & Future Research Rating: 3/5 Honest about some limitations but misses others; proposes useful but underspecified future research

VIII. WRITING QUALITY AND ACADEMIC STANDARDS

Rating: GOOD (4/5)

Strengths:

1. **Clear Structure:** Standard IMRAD format (Introduction, Methods, Results, Discussion) with logical progression
2. **Appropriate Citations:** 55 references, generally well-integrated, properly formatted
3. **Transparent Methods:** Section 3 provides detailed methodological information allowing replication
4. **Tables:** Well-formatted, comprehensive (Tables 1-5 present all necessary statistics)
5. **Honest Tone:** Author acknowledges contradictions and limitations rather than overselling findings

Weaknesses:

1. **Some Awkward Phrasing:**
 - "Financial distress requires redefinition" appears 4+ times almost verbatim
 - Occasional grammatical issues (minor)
2. **Inconsistent Terminology:**
 - "Financial distress," "financial viability," "financial condition" used somewhat interchangeably
 - Z-scores described as "high" (positive) but also "inflated" (problematic)
3. **Circular Arguments Not Flagged:** The paper doesn't explicitly acknowledge when it argues both sides (Z-scores are valid / Z-scores are invalid)
4. **Missing Standard Elements:**
 - No funding statement (disclosed as no funding, which is good)
 - No data availability statement beyond reference to public source
 - No ethics approval (arguably not needed for financial statements analysis)

Publication Venue: *Frontiers in Public Health* (Q2 journal, Impact Factor ~3-4) is an appropriate venue for this work. It's a health management/health policy journal, not a finance journal, which explains some of the methodological choices and theoretical limitations.

For publication in a Q1 finance or management journal, the paper would need:

- Stronger theoretical development
 - More sophisticated econometric methods
 - Causal identification strategy
 - Broader contribution beyond single-country description
-

FINAL SYNTHESIS AND RECOMMENDATIONS

Overall Assessment: C+/B- (70-75/100)

This paper addresses a relevant and underexplored question at the intersection of corporate finance and public healthcare management. It provides valuable empirical documentation of Greek public hospitals' financial characteristics and correctly identifies that traditional financial distress models fail in contexts of soft budget constraints and continuous government support. However, it falls short of making a significant theoretical or methodological contribution that would meet the standards of advanced corporate finance research.

What the Paper Does Well:

1. **Problem Identification:** Clearly articulates that traditional bankruptcy prediction models don't work for public entities
2. **Empirical Documentation:** Comprehensive data on 90 Greek hospitals with detailed financial information
3. **Honest Assessment:** Acknowledges that its primary analytical tool (Z-scores) is flawed in this context
4. **Policy Relevance:** Identifies real problems (payment delays, subsidy dependence, transferred distress) that matter for healthcare systems
5. **Gender Findings:** Provides evidence on female leadership effects in public sector context

Critical Weaknesses:

1. **Theoretical Underdevelopment:** Identifies need for new theory but doesn't develop it; limited engagement with corporate finance theory
2. **Methodological Limitations:** Cross-sectional design inappropriate for studying distress; bivariate analysis only; no causal inference
3. **Circular Reasoning:** Uses invalid measure (Z-scores) to test hypotheses, then argues measure is invalid
4. **Incomplete Contribution:** Calls for "redefining financial distress" but provides no redefinition
5. **Limited Generalizability:** Single country, single year, no comparative perspective

How This Paper Could Be Strengthened:

For Theoretical Contribution:

1. Develop a formal model of financial distress in soft budget constraint contexts
2. Specify what outcomes constitute "crisis" in public hospitals (beyond bankruptcy)
3. Apply existing corporate finance theory (pecking order, trade-off, agency) to explain observed patterns
4. Engage with institutional economics and public choice theory

For Methodological Rigor:

1. Collect panel data spanning multiple years
2. Implement multivariate regression with controls for confounds
3. Use difference-in-differences or other quasi-experimental designs to establish causality
4. Develop and validate an alternative distress measure specific to public hospitals

For Empirical Contribution:

1. Add comparative dimension (other countries, private hospitals, before/after reforms)
2. Link financial metrics to actual crisis outcomes (mergers, closures, interventions)
3. Examine mechanisms (why does gender matter? why does size matter?)

For Practical Impact:

1. Propose specific financial metrics for public hospital oversight
2. Design disclosure requirements for subsidy dependence and payment delays
3. Model policy interventions and predict their effects
4. Provide decision tools for government funders

Fit with Advanced Corporate Finance Course:

Appropriate uses in class:

- Case study of applying finance tools to non-standard contexts
- Discussion of institutional differences and model validity
- Example of when theory needs adaptation
- Illustration of research design challenges

Limitations for course use:

- Doesn't demonstrate state-of-the-art empirical methods
- Limited engagement with capital structure and governance theory
- No coverage of 40% of syllabus topics (M&A, real options, diversification)

Recommendation: This paper is suitable for a class session on "Applying Corporate Finance to Public Sector Organizations" or "Institutional Context and Financial Models," but should be supplemented with papers showing stronger theoretical and methodological approaches.

Bottom Line:

This is competent descriptive research that identifies an important problem and provides useful empirical documentation of a specific context. It makes a modest contribution to understanding public hospital finance and raises important questions about the applicability of private sector financial tools to public entities. However, it does not provide the theoretical development, methodological sophistication, or generalizable insights expected of research published in top-tier management or finance journals. It represents solid work for a health management journal but would need substantial development to meet standards for a Q1 finance outlet or to serve as an exemplar of advanced corporate finance research.